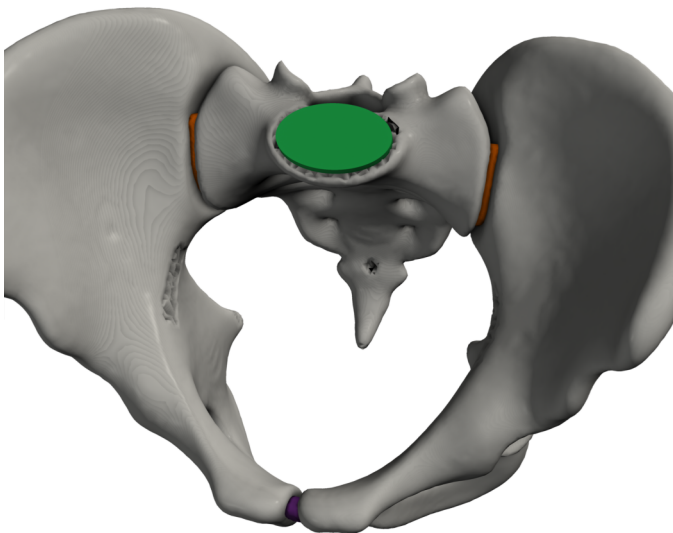
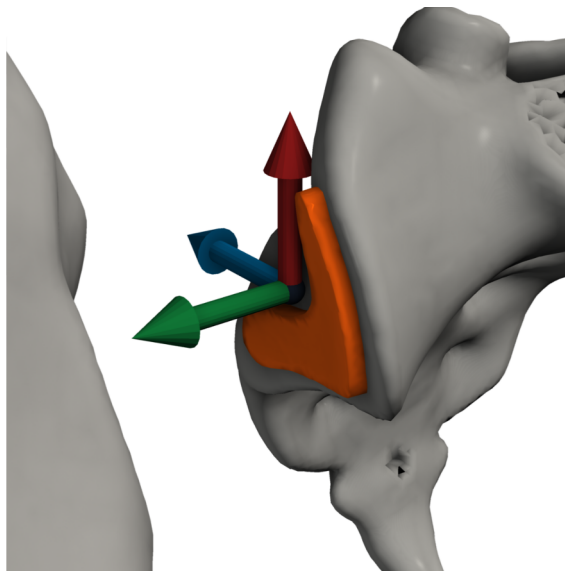


A. Pelvic assembly



Green: S1 constraint Orange: SIJ cartilage
Purple: pubic symphysis

B. SIJ articulation



Exploded view; arrows illustrate axis directions
Green: ML Blue: AP Red: CC

C. Joint-motion definitions

Reference

Unloaded subject anatomy

Rotations

Fixed-axis xyz: $R_z(\gamma)R_y(\beta)R_x(\alpha)$

Translations

Difference of fitted bone maps at the
sacral region-of-interest centroid

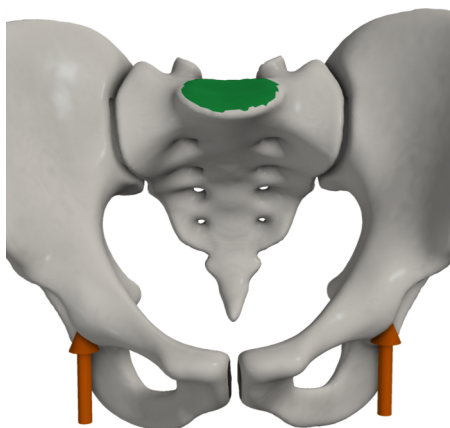
Reporting axes

Pelvis-aligned ML, AP and CC;
absolute components are compared

Bilateral asymmetry

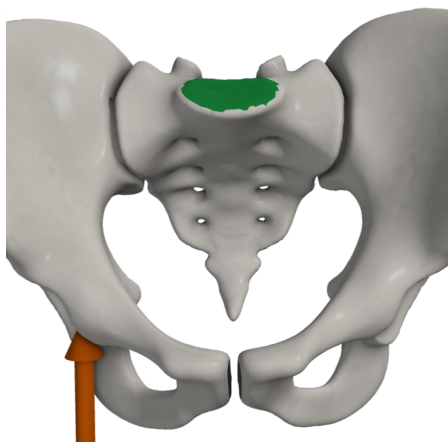
$$A_d = ||\mathbf{d}_L|| - ||\mathbf{d}_R||$$

D. SP2leg



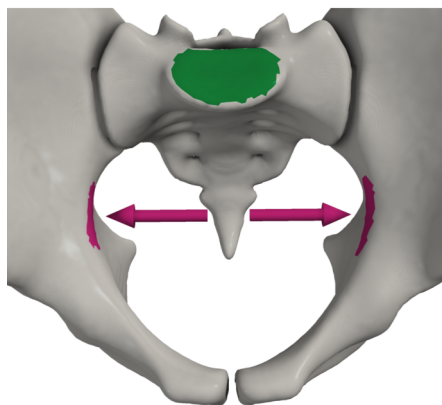
Bilateral acetabular load
400 N per side; 800 N total

E. SP1leg



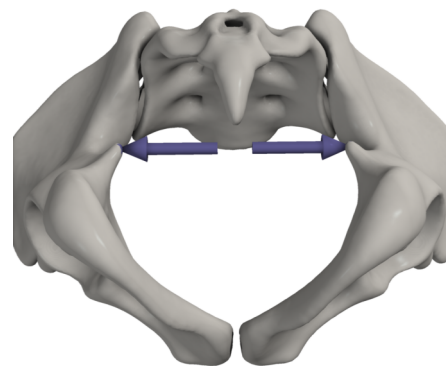
Right acetabular load
800 N total

F. LAB1



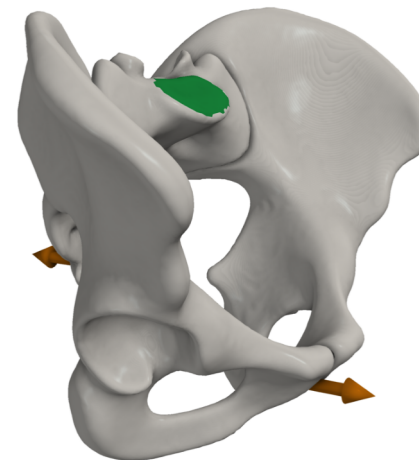
Internal ring force pair
Opposing ML forces: 400 N each

G. LAB2



Ischial force pair
Opposing ML forces: 400 N each

H. LAB3



Pubis-sacrococcygeal pair
Opposing AP forces: 400 N each